

ERBOT v0.2.0 — Integrated Benchmark Results

CORA · Affiliation Strings · Synthetic 10K · NCVR 5-Party · NCVR 10-Party

ERBOT v0.2.0

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Abstract

This document integrates the individual ERBOT benchmark results across five datasets: CORA (bibliographic deduplication), Affiliation Strings (free-text institution deduplication), Synthetic 10K (noisy unstructured text deduplication), NCVR 5-Party (two-source voter record linkage at two scales), and NCVR 10-Party (multi-source voter record linkage at 100K scale). It synthesises

per-dataset findings into cross-cutting conclusions about method reliability, blocking effectiveness, scalability limits, and recommended configurations for ERBOT v0.2.0.

1 Benchmark Overview

1.1 Datasets

Six experimental conditions across five benchmark datasets were evaluated.

ID	Dataset	Records	Fields	Gold pairs	Match rate*	Task
C	CORA Bibliographic	1,879	15	64,578	3.7%	Deduplication
A	Affiliation Strings	2,260	1	32,816	1.3%	Deduplication
S	Synthetic 10K (d10k)	10,000	1	8,705	0.017%	Deduplication
N5-s	NCVR 5-Party (small)	10,000	4	~880	0.018%	Linkage
N5-l	NCVR 5-Party (large)	100,000	4	~880	<0.001%	Linkage
N10	NCVR 10-Party	100,000	4	1,512	0.0003%	Linkage

Table 1: Benchmark conditions. *Match rate = gold pairs / $\binom{n}{2}$ (fraction of all possible record pairs that are true matches). CORA and Affiliation are relatively dense; single-field and linkage datasets are very sparse.

1.2 Pipeline Configuration Summary

ID	Blocking	Pairs	RR	Methods run	Runtime
C	Prefix-3 on title	137,707	92.2%	9 (incl. SVM)	≈59 min
A	Prefix-3 on affiliation	164,719	93.5%	9 (incl. SVM)	≈5.6 hr
S	Prefix-3 on text	885,301	98.2%	8 (SVM excluded ^a)	≈11 min
N5-s	Prefix-3 on surname	~10,600	≈100%	9 (incl. SVM)	<1 hr
N5-l	Prefix-3 on surname	1,059,972 ^b	≈100%	4 (graph-only ^c)	≈45 hr
N10	Prefix-3 on surname	5,275,664	99.9%	3 (graph-only ^c)	≈8 min

Table 2: ^a SVM killed after ≈20 hours (885K pairs, estimated 1–4 days to complete). ^b Truncated from 2,121,188 (pair budget cap exceeded). ^c hclust/pam/gc excluded: exceed R’s 65,536-observation distance-matrix limit or require 74.5 GiB dense allocation. N10 uses 5M pair budget vs. 2M for N5-l.

2 Per-Dataset Results

2.1 CORA Bibliographic Dataset (C)

Characteristics: 1,879 research paper citations with 15 fields; heavy noise (OCR errors, missing fields, formatting inconsistencies). Ground truth: 64,578 matched pairs.

Key points: hclust_avg (ARI = 0.746) is the best method on CORA, and the highest ARI result across all deduplication benchmarks. PAM (0.624) and SVM (0.543) follow. The **final** merge now correctly selects hclust_avg. With 15 discriminative fields, hierarchical and centroidal methods outperform the graph-based family (threshold-CC: 0.506).

2.2 Affiliation Strings Dataset (A)

Characteristics: 2,260 author affiliation strings, one noisy text field, extreme surface-form variability (abbreviations, language variants, reorderings). Ground truth: 32,816 matched pairs.

Method	ARI	NMI	B ³ -P	B ³ -R	B ³ -F	PF-F
hclust_avg	0.746	0.855	0.651	0.849	0.737	0.757
hclust_ward	0.524	0.827	0.661	0.748	0.702	0.542
pam	0.624	0.841	0.639	0.784	0.704	0.638
svm	0.543	0.849	0.630	0.817	0.711	0.566
threshold_cc	0.506	0.811	0.520	0.858	0.648	0.532
louvain	0.506	0.811	0.520	0.858	0.648	0.532
label_prop	0.506	0.811	0.520	0.858	0.648	0.532
leiden	0.000	0.686	1.000	0.065	0.122	0.000
gc [†]	0.000	0.000	0.040	1.000	0.077	0.076
<i>final</i>	0.746	0.855	0.651	0.849	0.737	0.757

Table 3: CORA performance (rerun 2026-03-26, merge bug fixed). [†]gc failed with dimension mismatch error.

Method	ARI	NMI	B ³ -P	B ³ -R	B ³ -F	PF-F
threshold_cc	0.076	0.688	0.395	0.550	0.460	—
louvain	0.076	0.688	0.394	0.550	0.459	—
label_prop	0.076	0.688	0.394	0.550	0.459	—
svm	0.035	0.818	0.977	0.159	0.273	—
leiden	0.000	0.819	1.000	0.146	0.255	—
hclust_avg	0.053	0.516	0.184	0.530	0.272	—
pam	0.053	0.516	0.184	0.530	0.272	—
hclust_ward	0.042	0.516	0.185	0.519	0.273	—
gc	0.000	0.000	0.007	1.000	0.014	—
<i>final</i>	0.076	0.688	0.395	0.550	0.460	—

Table 4: Affiliation Strings performance. PF columns omitted (in individual summary).

Key points: Overall performance is very low (best ARI = 0.076). SVM paradoxically underperforms threshold-CC (ARI 0.035 vs. 0.076): extreme precision (0.977) but near-zero recall (0.159) — it only merges the highest-confidence pairs. Primary bottleneck: prefix-3 blocking on a variable-opening-token field discards many true matches before any clustering is attempted. Final merge correctly selected threshold-CC.

2.3 Synthetic 10K Dataset (S)

Characteristics: 10,000 noisy unstructured text records, one field, corruptions via character substitution, token deletion, and reordering. Ground truth: 8,705 matched pairs.

Key points: Performance is the lowest of any single-condition run (best ARI = 0.023). An interesting inversion: Leiden has the highest NMI (0.914) and B³-F (0.426) but ARI = 0 due to over-splitting. Threshold-CC, Louvain, and Label Propagation tie at ARI = 0.023 with more balanced recall (0.519 vs. 0.271). SVM is infeasible. Blocking (98.2% reduction) is again the primary bottleneck.

2.4 NCVR 5-Party Dataset (N5)

Characteristics: North Carolina Voter Registration records, 4 fields (`givenname`, `surname`, `suburb`, `postcode`). The dataset file contains 5 parties; the current runs used parties 0 and 1 (two-source linkage), 20% overlap, 2 fields modified per duplicate. Run at two scales: 5K/party

Method	ARI	NMI	B ³ -P	B ³ -R	B ³ -F	PF-F
leiden	0.000	0.914	1.000	0.271	0.426	—
threshold_cc	0.023	0.778	0.334	0.519	0.406	—
louvain	0.023	0.778	0.334	0.519	0.406	—
label_prop	0.023	0.778	0.334	0.519	0.406	—
hclust_avg	0.002	0.371	0.019	0.573	0.036	—
pam	0.002	0.371	0.019	0.573	0.036	—
hclust_ward	0.001	0.347	0.019	0.690	0.037	—
gc	0.000	0.000	0.001	1.000	0.002	—
svm	<i>not run — estimated 1–4 days; killed after 20 hr</i>					
<i>final</i>	0.023	0.778	0.334	0.519	0.406	—

Table 5: Synthetic 10K performance. PF columns omitted (in individual summary).

(10K total, small) and 50K/party (100K total, large).

Scale	Method	ARI	NMI	B ³ -P	B ³ -R	B ³ -F	PF-F
Small	threshold_cc / louvain / label_prop / svm (tied)	0.917	0.975	1.000	0.929	0.963	0.923
	hclust_avg	0.276	0.713	0.537	0.929	0.681	0.364
	pam	0.158	0.672	0.537	0.786	0.638	0.258
	leiden	0.000	0.849	1.000	0.500	0.667	0.000
	gc	0.000	0.000	0.143	1.000	0.250	0.143
Large	threshold_cc	0.195	0.956	0.991	0.560	0.715	0.195
	louvain	0.133	0.947	0.933	0.560	0.700	0.134
	label_prop	0.133	0.947	0.933	0.560	0.700	0.134
	leiden	0.000	0.951	1.000	0.500	0.667	0.000
	hclust/pam/gc	<i>failed (65K limit / 74.5 GiB allocation)</i>					
<i>final (both)</i>		0.917 / 0.195	0.975 / 0.956	1.000 / 0.991	0.929 / 0.560	0.963 / 0.715	0.923 / 0.195

Table 6: NCVR 5-Party performance at small (10K records) and large (100K records) scale.

Key points: NCVR at small scale is the strongest result across all benchmarks (ARI = 0.917), benefiting from a 4-field structured profile and controlled synthetic noise. At large scale, ARI drops to 0.195 — not because the clustering degrades, but because the match density is 10× lower (880 true matches in 100K records vs. 10K records). Precision stays at 0.991; recall drops from 0.929 to 0.560. Four methods (hclust_avg, hclust_ward, pam, gc) fail at 100K scale and have been removed from the large-run script.

2.5 NCVR 10-Party Dataset (N10)

Characteristics: 100,000 NCVR records across 10 parties (10K per party); 4 structured fields; 20% overlap, 2-field modification. True cross-party pairs: 1,512 (match density 0.030%). Graph-only methods; 5M pair budget.

Key points: threshold_cc outperforms louvain/label_prop by a large margin (B³-F 0.900 vs. 0.769) in contrast to NCVR5-small where all three methods tied. At very low match density (1,512 true pairs in 100K records), threshold_cc’s strict cut-off suppresses spurious merges more effectively than community-detection methods. The 5M pair budget (vs. 2M in NCVR5-large) is a key difference: NCVR5-large budget truncation likely contributed to its lower recall (0.560), while NCVR10 with 5M pairs achieves near-perfect recall (0.991). Note that NCVR5-large and NCVR10 are not directly comparable (different party counts and true pair distributions) — this is directional evidence, not a controlled budget experiment. Total runtime was 8.2 minutes (vs. ≈45

Method	B ³ -P	B ³ -R	B ³ -F	Hom.	Com.	V-measure
threshold_cc	0.824	0.991	0.900	0.951	0.999	0.975
louvain	0.629	0.991	0.769	0.835	0.999	0.909
label_prop	0.629	0.991	0.769	0.835	0.999	0.909
<i>final</i>	0.824	0.991	0.900	0.951	0.999	0.975

Table 7: NCVR 10-Party performance (100K records). ARI, NMI, VI, and PairF not computed at this scale ($O(n^2)$ evaluation bottleneck). Hom. = Homogeneity; Com. = Completeness.

hours for NCVR5-large) — the improvement is entirely due to removing failed distance-matrix methods.

3 Cross-Dataset Analysis

3.1 Overall Performance Comparison

Dataset	Best ARI	Best B ³ -F	Best V	Best PF-F	Final B ³ -F	Final correct?	Best method
CORA	0.746	0.737	0.855	0.757	0.737	Yes	hclust_avg
Affiliation	0.076	0.460	0.688	—	0.460	Yes	threshold_cc
Syn10K	0.023	0.426	0.914	—	0.406	Yes	threshold_cc
NCVR5 (small)	0.917	0.963	0.975	0.923	0.963	Yes	threshold_cc
NCVR5 (large)	0.195	0.715	0.956	0.195	0.715	Yes	threshold_cc
NCVR10	—	0.900	0.975	—	0.900	Yes	threshold_cc

Table 8: Best performance per dataset/condition. “Final correct” = did the **best** merge strategy select the actual best method? NCVR10 ARI/PairF not available (evaluation bottleneck at $n = 100,000$). V = V-measure.

3.2 Method Consistency Across Datasets

Method	CORA	Aff	Syn10K	N5-s	N5-l	N10 [†]	Assessment
hclust_avg	0.746	0.053	0.002	0.276	fail	n/a	Best on CORA; weak elsewhere; fails >65K
threshold_cc	0.506	0.076	0.023	0.917	0.195	0.900 B³	Best or tied-best in all 6 conditions
louvain	0.506	0.076	0.023	0.917	0.133	0.769 B ³	Matches threshold-CC except NCVR-large
label_prop	0.506	0.076	0.023	0.917	0.133	0.769 B ³	Identical to louvain in all conditions
pam	0.624	0.053	0.002	0.158	fail	n/a	Strong on CORA; weak elsewhere; fails >65K
svm	0.543	0.035	n/a	0.917	n/a	n/a	Good on multi-field; infeasible at scale
hclust_ward	0.524	0.042	0.001	0.068	fail	n/a	Moderate on CORA; weak elsewhere; fails >65K
leiden	0.000	0.000	0.000	0.000	0.000	n/a	ARI = 0 in all conditions
gc	0.000	0.000	0.000	0.000	0.000	n/a	ARI = 0 / fails in all conditions

Table 9: Primary metric by method and condition (n/a = not run; fail = runtime error). CORA/Aff/Syn10K/N5 values are ARI; N10[†] values are B³-F (ARI unavailable at 100K scale).

Three clear groups emerge:

1. **Reliable graph methods:** `threshold_cc`, `louvain`, `label_prop` are best or joint-best in every condition. They converge to an identical solution in 4 of 5 ARI-comparable conditions. In the N10 (B^3 -F) run, `threshold_cc` pulls ahead of `louvain`/`label_prop` by 0.131 B^3 -F at very low match density. `threshold_cc` is the single most robust method and should be the ERBOT default.
2. **Conditionally useful:** SVM wins on CORA (ARI +0.037 over graph methods) where 15 fields provide rich training signal. On single-field or large datasets it is worse than `threshold-CC` or infeasible. The SVM viability boundary appears to be $\approx 200K$ candidate pairs (complete in reasonable time up to 164K pairs, fails at 885K).
3. **Never useful at defaults:** `leiden` (ARI = 0 across all 5 ARI-comparable conditions), `gc` (ARI = 0 across all 5 ARI-comparable conditions), `hclust_avg/ward`, `pam`. These methods either degenerate or fail at scale without parameter tuning. They should not appear in production runs without explicit tuning.

3.3 Blocking Effectiveness

Dataset	RR	Blocking key	Gold pairs	Bottleneck?	Notes
CORA	92.2%	Prefix-3 on <code>title</code>	64,578	Moderate	Title prefix usually consistent in citations
Affiliation	93.5%	Prefix-3 on <code>affiliation</code>	32,816	Severe	Variable opening tokens (“Dept. of”, “Univ. of”)
Syn10K	98.2%	Prefix-3 on <code>text</code>	8,705	Severe	Corrupted/reordered first tokens
NCVR5-small	$\approx 100\%$	Prefix-3 on <code>surname</code>	~ 880	Low	Surname prefix stable under 2-field modification
NCVR5-large	$\approx 100\%$	Prefix-3 on <code>surname</code>	~ 880	High*	Budget cap (2.12M $\rightarrow$ 2M) truncated true pairs; recall=0.560
NCVR10	99.9%	Prefix-3 on <code>surname</code>	1,512	Low	5M budget sufficient; recall=0.991; no truncation

Table 10: Blocking effectiveness per dataset. *NCVR5-large 2M budget was the primary cause of recall=0.560; raising to 5M (as in NCVR10) recovers recall=0.991.

Prefix-3 blocking on the primary text field is appropriate only when the field has a consistent, reliable prefix (as in bibliographic titles and personal surnames). It is a poor choice for free-text affiliation strings and synthetic noisy text, where many true match pairs have incompatible opening tokens. For structured records (NCVR), surname prefix is stable and the limiting factor is the pair budget, not prefix quality.

3.4 Scalability Profile

3.5 Performance Drivers

The following factors explain performance variation across conditions, in approximate order of impact:

1. **Field count and noise quality** (primary). NCVR with 4 structured fields and controlled noise achieves ARI = 0.917; the single-field datasets plateau at 0.023–0.076. CORA with 15

Method	Max viable records	Bottleneck	Alternative
hclust_avg/ward	<65,536	$O(n^2)$ distance matrix	graph methods
pam	<65,536	$O(n^2)$ distance matrix	graph methods
gc	<65,536	Dense matrix allocation	remove from defaults
svm	~200K pairs	Super-linear kernel computation	logistic regression, GBM
threshold_cc	$\geq 100,000$	Sparse graph ops	—
louvain	$\geq 100,000$	Sparse graph ops	—
label_prop	$\geq 100,000$	Sparse graph ops	—
leiden	$\geq 100,000$	Sparse graph ops	—

Table 11: Scalability limits by method.

fields but heavy uncontrolled noise (OCR errors, missing fields) achieves ARI = 0.746 — lower than NCVR despite having more fields. Field count helps when fields are informative; noise quality determines how discriminative each field is. Both factors drive performance jointly.

2. **Match density.** At the same 4-field NCVR configuration, match density decreases as scale grows: small (N5-s, ~880 pairs in 10K records) $\rightarrow$ large (N5-l, ~880 in 100K) drops ARI from 0.917 to 0.195. When true matches are rare, blocking recall failures and false-positive cluster merges are amplified.
3. **Blocking recall.** On single-field noisy text (Affiliation, Syn10K), prefix-3 blocking is too aggressive: many true match pairs have incompatible prefixes and are eliminated before any similarity or clustering computation. The measured performance is a lower bound for these datasets.
4. **Supervision signal.** SVM outperforms graph methods on CORA (+0.037 ARI) where 15 fields provide a rich multi-dimensional feature space. On single-field datasets the feature space is one-dimensional and SVM’s decision boundary collapses to a scalar threshold — functionally equivalent to threshold-CC but slower.
5. **Noise model.** Controlled 2-field corruption (NCVR) is easier than uncontrolled real-world errors (CORA citations) or extreme surface-form variation (Affiliation). Datasets with predictable noise are easier for all methods.

3.6 The final Merge Strategy

Dataset	Selected	Correct?	Final ARI	Final B ³ -F
CORA	hclust_avg	Yes	0.746	0.737
Affiliation	threshold_cc	Yes	0.076	0.460
Syn10K	threshold_cc	Yes	0.023	0.406
NCVR5-small	threshold_cc	Yes	0.917	0.963
NCVR5-large	threshold_cc	Yes	0.195	0.715
NCVR10	threshold_cc	Yes	—	0.900

Table 12: Final merge selection per condition. — = not computed (evaluation bottleneck). The merge bug was present in the original CORA run and fixed before subsequent runs.

The merge selection bug (originally exposed by CORA) has been fixed. All six conditions now correctly identify the best method. The CORA rerun (2026-03-26) confirms **final** = hclust_avg at ARI = 0.746 — the strongest ARI result across all benchmarks. NCVR10 achieves B³-F = 0.900 with threshold_cc, the highest B-cubed score of any large-scale (100K) condition.

4 Unified Key Findings

1. **Threshold-CC is the most reliable ERBOT method.** It is best or tied-best in all six experimental conditions, scales to $\geq 100\text{K}$ records, and is fast. It should be the primary default clustering method for all ERBOT runs. At very low match density (NCVR10, 0.030%), `threshold_cc` separates from `louvain/label_prop` by 0.131 B³-F points — its strict cut-off criterion matters more as true match pairs become rare.
2. **Louvain and Label Propagation are robust second choices.** They produce results identical to threshold-CC in 4 of 5 ARI-comparable conditions. At very large scale with low match density (NCVR5-large, NCVR10) they underperform threshold-CC. All three graph methods are worth running together since their agreement is a strong quality signal.
3. **Leiden and GC should be excluded from default runs.** Both produce $\text{ARI} = 0$ in every condition where ARI was available without parameter tuning. Leiden over-partitions (too-high resolution leading to near-zero recall) and GC over-merges (degenerate fallback caused by an unresolved dimension mismatch bug). Until tuned per-dataset, they add noise to the method comparison and inflate runtime.
4. **hclust_avg is the best method on CORA (ARI = 0.746), but does not generalise to other datasets.** On CORA’s dense 15-field similarity matrix, average-link hierarchical clustering finds an excellent partition. On single-field datasets or at $>65\text{K}$ records it fails or degenerates. SVM ($\text{ARI} = 0.543$ on CORA) is more portable: it also ties for best on NCVR-small, though it is infeasible above $\approx 200\text{K}$ candidate pairs.
5. **The final merge is working correctly across all datasets.** The original CORA run had a merge selection bug that picked the worst method. After fixing and rerunning (2026-03-26), CORA `final` correctly selects `hclust_avg` at $\text{ARI} = 0.746$. All six conditions now report correct `final` results.
6. **Blocking is the primary performance bottleneck on single-field and noisy datasets.** Prefix-3 blocking is well-suited to bibliographic titles and personal surnames, but discards large fractions of true matches on affiliation strings and synthetic noisy text. Pair completeness (fraction of gold pairs retained after blocking) is not currently measured and should be added to Stage 3 diagnostics.
7. **Distance-matrix methods (hclust, pam, gc) are not viable at scale.** They fail hard at $>65\text{K}$ records due to R’s memory constraints. They should be gated behind a record-count check in `er_cluster_all()` and automatically excluded when $n > 65000$.
8. **Performance correlates strongly with field count.** NCVR (4 fields, controlled noise) $\gg$ CORA (15 fields, uncontrolled noise) $\gg$ Affiliation/Syn10K (1 field). Every additional discriminative field substantially reduces the difficulty of the clustering task.
9. **Pair budget is the dominant NCVR performance lever at scale.** NCVR5-large with a 2M budget achieves recall 0.560; NCVR10 with a 5M budget achieves recall 0.991 at the same $n = 100,000$ scale. Raising the budget from 2M to 5M recovers 0.431 recall points. Budget should scale with record count.
10. **Equal weights are reasonable but not optimal.** CORA has 15 fields of highly varying quality; supervised weight learning would likely add +0.05–0.10 ARI for SVM and threshold-CC. For NCVR (4 balanced fields) and single-field datasets, weight learning provides no benefit.
11. **Evaluation scalability is a bottleneck at $n \geq 100,000$.** ARI, NMI, VI, and PairF could not be computed for NCVR10 due to $O(n^2)$ loops in `er_bcubed()` and contingency-table routines. B-cubed and V-measure completed. This should be fixed with vectorised table lookups before any further large-scale benchmarking.

5 Unified Recommendations

Priority	Action
High	Fix $O(n^2)$ evaluation bottleneck in <code>er_bcubed()</code> and the contingency-table routines. Replace inner <code>which()</code> loops with vectorised table lookups so ARI, NMI, VI, and PairF can be computed at $n = 100,000$.
High	Gate distance-matrix methods behind a record-count check. In <code>er_cluster_all()</code> , automatically skip <code>hclust_avg</code> , <code>hclust_ward</code> , <code>pam</code> , and <code>gc</code> when $n > 65000$ rather than attempting and failing.
High	Add pair completeness to Stage 3 output. Report the fraction of ground-truth pairs retained after blocking. Currently all recall figures are lower bounds.
High	Remove leiden and gc from default method lists , or add a degenerate-detection guard that flags $ARI = 0$ results and excludes them from the <code>final</code> merge comparison.
Medium	Replace SVM with logistic regression or gradient boosting for datasets with $>200K$ candidate pairs. Provides near-linear scalability with similar accuracy in the multi-field setting.
Medium	Improve blocking for single-field and free-text datasets. For affiliation strings: sorted neighbourhood (SN) blocking or token-set Jaccard. For synthetic noisy text: SN blocking on the full field or n-gram shingling.
Medium	Set pair budget to at least $5 \times N$ for structured linkage tasks. NCVR10 demonstrated that a 5M budget at 100K records achieves $\text{recall}=0.991$, while the old 2M budget caused $\text{recall}=0.560$. Make the budget adaptive or scale with record count.
Medium	Tune Leiden resolution parameter per dataset. Default resolution is consistently too high; try 0.05–0.2. Until tuned, exclude Leiden from production comparisons.
Medium	Fix GC dimension mismatch bug. GC silently falls back to a degenerate solution on all datasets where it was run due to an unresolved interface error. Either fix or remove from the method list.
Low	Implement supervised weight learning for multi-field datasets. ARI-based weight optimisation on CORA expected to gain +0.05–0.10 ARI over equal weights.
Low	Use all 5 parties in the NCVR5 dataset (currently only parties 0 and 1 are used) to assess 5-source linkage under the full NCVR5 design. The 10-party design (NCVR10) has already been run; this fills the gap at intermediate party count.

6 Conclusions

ERBOT v0.2.0 has been evaluated across five benchmark datasets (six conditions) spanning a wide range of ER task types and sizes. The results establish a clear picture of what works, what scales, and what needs fixing.

What works reliably: Threshold-CC (and by extension Louvain and Label Propagation) is a robust, scalable default that performs best or jointly best in every condition. At very low match density (NCVR10, 0.030%), `threshold_cc` pulls 0.131 $B^3\text{-F}$ ahead of community-detection methods, making it even more valuable at large scale. The `final` merge strategy correctly identifies the best method in all six conditions (the CORA merge bug is now fixed).

What the numbers mean in context: The best achievable ARI varies enormously by task — from 0.917 on NCVR5-small (controlled noise, 4 fields) and 0.746 on CORA (15 fields, rich signal) to 0.023 on Syn10K (noisy single field). NCVR10 ($B^3\text{-F} = 0.900$) is the strongest large-scale result and shows that structured 4-field linkage is tractable at 100K records when the pair budget is sized appropriately. These differences reflect fundamentally different task difficulties, not ERBOT’s limitations.

What needs immediate attention: Three issues stand out. First, evaluation scalability: ARI,

NMI, VI, and PairF could not be computed for NCVR10 due to $O(n^2)$ inner loops — fixing this is a prerequisite for further large-scale benchmarking. Second, the four degenerate methods (leiden, gc, hclust/pam at scale) generate runtime failures and misleading zero-ARI entries in every result table — they should be gated or removed. Third, pair completeness at Stage 3 is unmeasured, meaning all recall figures are lower bounds.

Direction: The most impactful improvements would come from fixing evaluation scalability (to fully characterise NCVR10 and future large-scale runs), better blocking for free-text fields (Affiliation, Syn10K), and a scalable replacement for SVM. The NCVR10 result confirms that graph-only ERBOT runs at 100K scale are fast (<10 minutes) and accurate ($B^3\text{-F} \approx 0.90$) when the pair budget is adequate.

7 Output Files Reference

Path	Contents
res_cora/CORA_RESULTS_SUMMARY.pdf	Full CORA individual summary
res_cora/er_performance.csv	CORA metric table
res_aff/AFF_RESULTS_SUMMARY.pdf	Full Affiliation individual summary
res_aff/er_performance.csv	Affiliation metric table
res_syn10k/SYN10K_RESULTS_SUMMARY.pdf	Full Syn10K individual summary
res_syn10k/er_performance.csv	Syn10K metric table
res_ncvr5/NCVR5_RESULTS_SUMMARY.pdf	Full NCVR5 individual summary
res_ncvr5/small/er_performance.csv	NCVR5-small metric table
res_ncvr5/large/er_performance.csv	NCVR5-large metric table
res_ncvr10/NCVR10_RESULTS_SUMMARY.pdf	Full NCVR10 individual summary
res_ncvr10/10k/er_performance.csv	NCVR10 metric table
ERBOT_INTEGRATED_SUMMARY.pdf	This document